

INTERNAL TECH MEMO: Eko Learn Scalable Onboarding Architecture

Date: June 24, 2026

Subject: Technical Specification for Decentralized Data Ingestion and Secure Identity Provisioning

1. Executive Summary: The "Zero-Visit" Ingestion Philosophy

To achieve rapid, cost-effective penetration across Lagos State, Eko Learn is transitioning to a **decentralized, principal-led onboarding model**. This "Zero-Visit" approach eliminates the need for physical field operations by shifting the burden of data collection to the institutional edge while centralizing validation, processing, and security at the Lumia OS Tech HQ. This architecture ensures high-fidelity data ingestion and robust identity management without the overhead of on-site manual entry.

2. Technical Ingestion Pipeline

The ingestion pipeline is designed to transform unstructured physical media into structured, machine-readable datasets through a four-stage process:

- **Standardized Digitization:** Principals are provided with high-contrast, structured clear-paper templates. These templates utilize anchor points to ensure standardized field mapping for (Name, Class, Gender, Age, Phone Number).
- **Mobile-Optimized Capture:** The secure school portal features an integrated camera interface that performs **real-time image preprocessing** (e.g., perspective correction, deskewing, and adaptive thresholding) to optimize the image for high-accuracy recognition before upload.
- **Advanced OCR Engine:** The backend leverages a specialized OCR pipeline configured to handle handwriting variations. It will utilize confidence scoring to identify high-risk fields.
- **Closed-Loop Validation:**
 - **Automated Flags:** Any data point failing the confidence threshold (e.g., ambiguous characters or low-resolution image patches) triggers an automated notification to the principal's portal with specific instructions for rectification.
 - **Schema Enforcement:** Final ingestion into the database is contingent upon successful schema validation, ensuring all required fields are present and correctly formatted.

3. Identity and Security Architecture

To ensure the integrity of student records, we are implementing a multi-factor identity framework designed for the specific constraints of the Lagos educational environment:

- **Dynamic Identity Generation:** Upon successful data ingestion, the system auto-provisions a unique, permanent **Lumia Student ID (LSID)**.
- **Verification Trigger:** The system initiates an automated SMS gateway sequence to dispatch the LSID and verification prompts to the registered contact number.
- **Two-Factor Authentication (2FA) Flow:**
 - **Primary Identifier:** The LSID.
 - **Secondary Verification:** A time-sensitive, phone-based **One-Time Password (OTP)**.
 - **Session Security:** Access is strictly gated by the OTP challenge, preventing unauthorized account impersonation and ensuring that account ownership is tied to the verifiable, registered contact number.
- **Mastery Tracking:** The LSID serves as a immutable primary key in our database, linking all historical mastery data, academic progress, and assessment outcomes to a single, persistent identity profile.

4. Privacy, Compliance, and Data Governance

In alignment with the **Nigeria Data Protection Regulation (NDPR)**, we are committed to a "Privacy by Design" posture:

- **Explicit Consent & Transparency:** At the initial login, the platform will present a mandatory, simplified NDPR-compliant privacy disclosure. Access to the dashboard is contingent upon the user/guardian acknowledging that data is strictly for educational institutional purposes.
- **PII Minimization (Zero-PII Policy):**
 - **Data Isolation:** The database will operate on an air-gapped logic relative to external government networks to prevent lateral data leakage.
 - **Strict Scope:** Only essential Personally Identifiable Information (PII) is persisted. Non-essential fields are purged post-processing, and all PII is encrypted at rest using AES-256 standards.
- **Auditability:** Every transaction from image upload to OTP validation—is logged in a tamper-evident audit trail to ensure accountability and regulatory readiness.

5. Implementation Roadmap: Team Deliverables

Team	Focus Area	Critical Deliverables

Frontend	Portal & Auth	Build the Mobile Submission Portal (with camera integration) and the high-security OTP login module.
Backend/AI	Pipeline Ops	Refine OCR confidence scoring; implement logic for automated "re-submit" triggers to schools.
Infrastructure	Scale & Latency	Provision high-concurrency SMS gateways; implement auto-scaling for the ingestion API to manage peak-hour load.

Note to Engineering Teams: Given the variability of school-level connectivity in Lagos, the frontend must prioritize "resumable uploads" to prevent data loss during network instability. All API endpoints must be hardened against common vulnerabilities to ensure zero-access to the student PII database.